



Huang Bai Jian Pi decoction alleviates diarrhea and represses inflammatory injury via PI3K/Akt/NF- κ B pathway: *In vivo* and *in vitro* studies

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ABSTRACT

Ethnopharmacological relevance: Huang Bai Jian Pi (HBJP) decoction, a Chinese herbal formula based on the Pulsatilla decoction (PD) and Si Junzi decoction, is efficacy to treat clinical diarrhea in calves.

Aim of the study: The mechanism of HBJP decoction to treat calf diarrhea remains unclear. This study was to investigate the therapeutic effect and anti-inflammatory mechanism of HBJP decoction on diarrhea in rats.

Materials and methods: Thirty-six Sprague Dawley rats were randomly divided into control group, model group, PD group and three treated groups with HBJP decoction. The diarrheal model in rats was established by multiple factors including high-sugar and fat diet, high temperature and dampness environment, biological pathogenic factors. The diarrheal animals were treated with HBJP decoction or PD for 5 days. The inflammatory model of the intestinal epithelioid cell line 6 (IEC-6) was induced by TNF- α . The clinical symptoms, blood routine and biochemistry parameters, histopathology of main organs were detected. The proteins associated with PI3K/Akt/NF- κ B pathway and the expression levels of cytokines associated with inflammation were detected *in vivo* and *in vitro* by Western blot and ELISA.

Results: The model rats showed obvious diarrheal symptoms, and the obvious systemic inflammatory response accompanied with abnormal change in blood routine, biochemistry parameters and histopathology. HBJP decoction alleviated obviously the clinical symptoms, and pathological changes of the liver, colon and lung, and abnormal blood routine and biochemistry indexes in rats. The expression of P-PI3K, P-Akt, P-NF- κ B, IL-1 β , IL-6 was significantly increased, and the expression of IL-10 was markedly decreased in diarrheal rats and IEC-6 with inflammation. HBJP decoction significantly inhibited the PI3K/AKT/NF- κ B signal pathway and adjusted the expression of these inflammatory cytokines.

Conclusions: The finding suggested that HBJP decoction alleviate the inflammation in diarrhea through inhibiting the PI3K/Akt/NF- κ B signal pathway, which provides scientific evidences for the clinical application of HBJP decoction in diarrhea.

1. Introduction

Diarrhea is one of the most common diseases in calves and still a major cause of economic loss to cattle industry worldwide. Diarrhea is usually caused by infectious and/or non-infectious factors (Caffarena et al., 2021). Multiple entero-pathogens including cryptosporidium, bovine rotavirus, coronaviruses and *Escherichia coli* (*E. coli*) are the common causes of infectious diarrhea in calves (Brainard et al., 2021;

Dall Agnol et al., 2021), and *E. coli* is the most common cause of infectious diarrhea in calves (Kolenda et al., 2015), therefore it was often used to construct diarrhea-associated models in animals (Hirudkar et al., 2020; Nyholm et al., 2015). Climatic factors including abnormal temperature and humidity also play an inducing role in diarrheal diseases. High temperature and high dampness are more conducive to the growth of bacteria and other pathogenic microorganisms, which easily induce the diarrhea (Aik et al., 2020). In addition, when animals were lived in high temperature and high humidity environments for a long time, their

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Abbreviations

HBJP	Huang bai jian pi
TCM	traditional Chinese medicine
PD	Pulsatilla decoction
PI3K	phosphatidylinositol 3-kinase
Akt	Protein kinase B
NF-κB	nuclear factor kappa B
IL	interleukin
IEC-6	intestinal epithelioid cell line 6
TNF	tumor necrosis factor
ELISA	enzyme-linked immunosorbent assay
UPLC-QE-MS	Ultra performance liquid chromatography-quadrupole electrostatic field orbital well-mass spectrometry
HE	hematoxylin and eosin
PVDF	polyvinylidene difluoride

MONO	monocyte
MPV	mean platelet volume
PCT	Platelet cubic measure distributing
IRF	immature reticulocyte fraction
NEUT	neutrophilic granulocyte
RET	reticulocyte count
WBC	white blood cell count
AMY	Serum amylase
GLOB	globulin
GLU	glucose
NA	sodium ion
ALB	albumin
BUN	blood urea nitrogen
CA	serum calcium
PHOS	phosphorus
SD	Sprague-Dawley
PLT	platelet.

intestinal flora and intestinal barriers were easily damaged by *E. coli* or other pathogens (Chen et al., 2019). Thus, the environment of high temperature and dampness is inducing cofactor of diarrhea in animals (Li et al., 2021). Non-infectious diarrhea has been associated with poor feeding conditions, in which high-sugar diets was reported as a common inducer to diarrhea (Hua et al., 2020). Studies had already shown that high-sugar diets can cause diarrhea in rats by exacerbating intestinal inflammation and inhibiting immune response (Zuo et al., 2021). High fat diet is another non-infectious cause to increase diarrheal disease (Francesio et al., 2020), and a growing number of studies showed that obesity is positively associated with diarrhea (Jose et al., 2021). Improper diet is also a major factor in diarrhea, such as starvation, which can lead to intestinal barrier defects and diarrhea (Kelly, 2021).

Currently, a variety of antibiotics including gentamicin, enrofloxacin, penicillin and streptomycin are commonly used for treatment of various type of diarrhea, but some risk factors including antibiotics residues and resistance in some bacteria were produced by misusing or overusing of antibiotics in herds. There is a wide variety of antibiotics available to treat diarrhea, but because most of these antibiotics lack on-label indications for treatment of diarrhea, it is difficult for veterinarians to use them properly (Olson et al., 2019). Therefore, it is necessary to develop safe and effective alternative therapeutic strategies for diarrhea. Many herbs in TCM have been used to treat diarrhea for a long history, and had remarkable clinical efficacy (Lv et al., 2017). HBJP decoction is a modified Chinese herbal formula based on the PD and Si Junzi decoction with an effective rate of 85.9% in treating calf diarrhea (supplementary material 1). HBJP decoction consists of eleven herbs including radix pulsatillae, rhizoma coptidis, pomegranate rind, rhizoma atractylodis macrocephalae, radix sanguisorbae, radix codonopsis, rhizoma atractylodis, poria, cortex magnoliae officinalis, white peony root and radix glycyrrhizae. Rhizoma coptidis, pomegranate rind, rhizoma atractylodis macrocephalae and radix sanguisorbae are main components in classic prescription of TCM for treating diarrhea in human being and animals (Shen et al., 2021; Tjong et al., 2011; Wang et al., 2019), and radix pulsatillae, radix sanguisorbae, cortex magnoliae officinalis, pomegranate rind and rhizoma coptidis have been proven to alleviate inflammation *in vitro* and *in vivo* tests (Chiu et al., 2021; Du et al., 2019; Fujii et al., 2017; Kang et al., 2019; Yasueda et al., 2020).

Diarrhea is often accompanied with enteritis, and PI3K/Akt signal pathway is involved in the pathogenesis of diarrhea-related diseases by regulating various inflammatory factors (Wu et al., 2013). Cytokines, viruses, and other bio-factors can lead to phosphorylation of PI3K, which continues to activate Akt by phosphorylation, and then the activated Akt activates NF-κB. The evidences shown that PI3K/Akt/NF-κB signal pathway participate in the pathogenesis of diarrhea-related

diseases by regulating various inflammatory factors (Rahmani et al., 2020). Therefore, the PI3K/Akt/NF-κB signal pathway becomes a target for treatment of diarrhea or screening drugs. Whether HBJP decoction could exert anti-inflammatory activity to alleviate diarrhea via PI3K/Akt/NF-κB signal remains unknown.

In this study, the main components of HBJP decoction were identified by Ultra performance liquid chromatography-quadrupole electrostatic field orbital well-mass spectrometry (UPLC-QE-MS). The diarrheal model in rat was established by multiple factors, and the effects of HBJP decoction on diarrheal rats and anti-inflammatory mechanisms were investigated. The finding suggested that HBJP decoction alleviates diarrhea and represses inflammatory injury through inhibiting the PI3K/Akt/NF-κB signal pathway.

2. Materials and methods

2.1. Preparation of HBJP decoction

The HBJP decoction is based on the PD and Si Junzi decoction, and contains 11 herbs. The specific information and composition ratios of these herbs are shown in Table 1. The herbs were purchased from the Yellow River Market of Medicinal Materials in Lanzhou, Gansu, China. Herbs were authenticated by Prof. Jianxi Li of the Lanzhou Institute of Husbandry and Pharmaceutical Sciences of Chinese Academy of Agricultural Sciences. All herbs were identified by Fuxing Hou Biopharmaceutical Technology Co., Ltd. (Gansu, China), and all of them were in accordance with the provisions of Chinese Pharmacopoeia.

The mixture of 11 herbs was soaked for 1 h in 10-fold (v/w) the amount of distilled water, followed by decoction for 1.5 h. The filtrate was collected, and the residues were subjected to decoction twice by the same extraction proposal. All filtrate was mixed and condensed to 1.00 g/mL (crude drug). The density and pH of the decoction were 1.05 g/mL and 4.43, respectively.

2.2. UPLC-QE-MS analysis of HBJP decoction

The HBJP decoction was vortexed for 30 s to thoroughly mix, and subsequently centrifuged at 12,000 rpm at 4 °C for 10 min. The 300 μL supernatant from each sample was transferred to a new Eppendorf tube, and 1000 μL extracting solution (Methanol: Distilled water = 4:1, v/v, the concentration of internal standard is 10 μg/mL) was added. Then, the samples were vortexed for 30 s, and sonicated in an ice water bath for 10 min. The samples were placed at −40 °C for 1 h, and centrifuged at 12,000 rpm for 10 min. Finally, the supernatant was filtered using a 0.22 μm filter membrane and stored at −80 °C until it was used for

Table 1

Herb composition of the HBJP decoction.

Latin name	Chinese name	English name	Batch number	Plant part used	Position	Composition (%)
<i>Pulsatilla chinensis</i> (Bunge) Regel	Baitouweng	Radix pulsatillae	Y1810042	Root	Jilin, China	16
<i>Coptis chinensis</i> Franch.	Huanglian	Rhizoma coptidis	Y1810041	Rhizome	Sichuan, China	8
<i>Atractylodes lancea</i> (Thunb.) DC.	Cangzhu	Rhizoma atractylodis	Y1810046	Rhizome	Inner Mongolia, China	8
<i>Atractylodes macrocephala</i> Koidz.	Baizhu	Rhizoma atractylodis macrocephalae	Y1810050	Rhizome	Hubei, China	16
<i>Poria cocos</i> (Schw.) Wolf.	Fuling	Poria cocos	Y1810044	Sclerotium	Yunnan, China	12
<i>Sanguisorba officinalis</i> Linn.	Diyu	Radix sanguisorbae	Y1810043	Root	Gansu, China	8
<i>Magnolia officinalis</i> Rehd. et Wils.	Houpu	Magnolia officinalis	Y1810047	Cortex	Sichuan, China	8
<i>Codonopsis pilosula</i> (Franch.) Nannf.	Dangshen	Radix codonopsis	Y1810051	Root	Gansu, China	8
<i>Pericarpium Granati</i>	Shiliupi	Pomegranate rind	Y1810045	Cortex	Shannxi, China	8
<i>Radix Paeoniae Alba</i>	Baishao	White peony root	Y1810049	Root	Anhui, China	4
<i>Glycyrrhiza uralensis</i> Fisch.	Gancao	Radix glycyrrhizae	Y1810048	Rhizome	Gansu, China	4

Note: the batch number is issued by Fuxing Hou Bio-pharmaceutical Technology Co., LTD after the herbs have been tested.

UPLC-QE-MS analysis.

UPLC-QE-MS analysis was performed on a 1290 UPLC system (Agilent Technologies Inc., USA) with a Waters UPLC BEH C18 column (1.7 μ m 2.1 * 100 mm). The injection volume of sample was set at 5 μ L. The mobile phase consisted of 0.1% formic acid in water (A) and 0.1% formic acid in acetonitrile (B). The flow rate was set at 0.4 mL/min and a multi-step linear gradient elution was applied (0–3.5 min, 95–85% A; 3.5–6 min, 85–70% A; 6–6.5 min, 70–70% A; 6.5–12 min, 70–30% A; 12–12.5 min, 30–30% A; 12.5–18 min, 30–0% A; 18–25 min, 0–0% A; 25–26 min, 0–95% A; 26–30 min, 95% A). A Q Exactive Focus mass spectrometer (Thermo Fisher Scientific, USA) coupled with Xcalibur software was employed to obtain the MS and MS/MS data based on the IDA acquisition mode. Sheath gas flow rate: 45 Arb; Aux gas flow rate: 15 Arb; Capillary temperature: 400 °C; Full ms resolution: 70,000; MS/MS resolution: 17,500; Collision energy: 15/30/45 in NCE mode; Spray Voltage: 4.0 kV (positive) or –3.6 kV (negative). The mass spectrometry data were identified using the secondary mass spectrometry database provided by Shanghai BIOTREE Biotechnology Co., Ltd., and matched with the corresponding methods.

2.3. Establishment of the rat model and groupings

Thirty-six Sprague Dawley rats, weighing 180–200 g, were purchased from the Experimental Animal Centre of Lanzhou Veterinary Research Institute, CAAS (Gansu, China, SCXK[Gan]2015-0001). All rats were acclimated for 1 week before the experiment. After a 7-d adaptation period, 6 out of 36 rats were randomly placed into the control group, and remaining 30 rats were used to create diarrheal models according to an established modified method (Hua et al., 2020; Yao et al., 2017). Rats were given 30% (V/V) honey water throughout the whole modelling process, and were fed with lard (4 mL/200 g body weight) and fasted every other day for 10 consecutive days. During the following 5 d, the rats were given 56-degree of liquor (2 mL/200 g body weight) every morning and exposed to a temperature and humidity environment of 31 ± 1 °C and $93 \pm 1\%$ for 8 h. Then in the following 2 d, rats were intraperitoneally injected with a 1.0×10^9 CFU/mL *E. coli* suspension (3 mL/200 g body weight) in the morning and fed with 0.1 g/mL of senna (2 mL/200 g body weight) in the afternoon. The *E. coli* (BNCC125787) used in the experiments was purchased from Beina Bio, Beijing, China. After model establishment process was finished, the rats were randomly divided into 5 groups, including model group, HBJP decoction high-dose group (H-HBJP group), HBJP decoction medium-dose group (M-HBJP group), HBJP decoction low-dose group (L-HBJP group) and Pulsatilla decoction group (PD group). The rats in model group were sacrificed, and the rats of other groups were administered the designed dose of drugs by gavage for 5 consecutive days, H-HBJP group treated with HBJP at 30 g/kg body weight, M-HBJP group treated with HBJP at 15 g/kg body weight, L-HBJP group treated with HBJP at 7.5 g/kg body weight and PD group treated with PD at 15 g/kg body weight. Rats in control group were administered the same volume of 0.9% saline for 5

consecutive days. The experimental doses of HBJP decoction were designed by converting the body surface area with reference to the clinical dose. In brief, HBJP decoction at a dose of 200 g per head has a therapeutic effect on calves with diarrhea, and the body weight and the body surface area of calf is about 50 kg and 1.304 m² (Hu et al., 1989), and the body weight and the body surface area of rats is about 200 g and 0.031 m². Thus, the effective dose of HBJP decoction in clinic was about 30 g/kg in rats.

At the end of the experiment, rats were sacrificed by intraperitoneal injection of 20% ethyl carbamate (1 mL/100 g body weight). In animal test, the blinding method was used, and the test groups were named numbers, and the different doses of drugs were also named letters, the research team has access to the real meaning of numbers and letters until the trial is complete. The animal experiment procedures were approved by the Ethics Committee of Lanzhou Institute of Husbandry and Pharmaceutical Sciences, CAAS (SYXK[Gan]2019-0002).

2.4. Measurement of clinical symptoms

Behaviour and clinical signs of rats, including mental, activity, hair, crissum and feces were recorded during the experiment. All rats were evaluated by semi-quantitative scoring according to their clinical symptoms (Table 2). The diarrheal index scores (diarrhea index = loose stools ratio * loose stools grade) of rats were recorded at the beginning of the experiment, from establishment of the model to the end of treatment (Lu et al., 2020). Body weight and body temperature were measured at the beginning of modelling (day 1), during modelling (day 10), after modelling (day 17), and after treatment with drugs (day 23), respectively.

2.5. Histopathological assays

Heart, liver, lung, kidney, spleen and colon were collected immediately after rats were euthanized. Organs were cut into small pieces and fixed in 4% paraformaldehyde for 2 weeks. The tissues were sectioned and stained with hematoxylin and eosin (HE) using the routine protocols. Three HE-stained sections were examined per specimen under a microscope (BX43F, Olympus, Japan).

2.6. Assays of blood routine and biochemistry

The blood samples were collected from the heart in vacuum tubes with EDTA anticoagulant for testing of blood routines, and in vacuum tubes without anticoagulant to serum collecting for testing of blood biochemical indexes respectively. Blood routines were tested by ProCyte Dx hematology analyzer (IDEXX, USA). Biochemical indexes of serum were tested by VS2 automatic biochemical analyzer (Zoetis, USA).

Table 2

Semi-quantitative assessment of clinical Manifestations of diarrheal rats.

score	Mental state	Activity state	growing status	Color of stools	Morphology of stools	The crissum	The hairs
0	lively and active	lively and active	gain weight	black	formed manure	white and clean	clean and lustrous
1	reduced voluntary activity	reduced voluntary activity	weight loss	brown	loose stools like water	dirty and red	yellow and lusterless
2	compulsive lying lazy to move	compulsive lying and lazy to move	slim down	yellow	mucus stool or bloody purulent stool	filthy, redness and swelling	Withered and sparsely shed

2.7. Cell culture and treatment

The IEC-6 was purchased from Beina Biological Company, China. Cells were cultured in high glucose DMEM medium (HyClone, USA) containing 10% fetal bovine serum and 1% penicillin and streptomycin (Gibco, USA), and incubated in a humidified atmosphere at 37 °C and 5% CO₂.

A cell counting kit-8 (CCK-8) was used to detect the effects of TNF- α , and the HBJP decoction on the proliferation of IEC-6 to determine the doses of these two factors. In brief, the IEC-6 were seeded on 96-multi-well plates at a density of 1×10^4 cells per well for culture. After cells were treated with different concentrations of TNF- α , and HBJP decoction for 24 h, 10 μ L CCK-8 was added and the preparation was cultured in the incubator for 1.5 h. Then, absorbance at a 450 nm was detected, and cell viability rates were calculated according to the absorbance value. Finally, the rational doses of these two treatments were determined.

To verify the roles of the PI3K/Akt/NF- κ B signal pathway in the anti-inflammatory of the HBJP decoction, the IEC-6 inflammatory model was established using TNF- α . LY294002 (Med Chem Express LLC, Shanghai, China) was used as an inhibitor of PI3K/Akt/NF- κ B signal pathway, and Recilisib (Med Chem Express LLC, Shanghai, China) was used as an activator of the same pathway. The experiment was divided into a control group, a model group (TNF- α 10 ng/mL), a HBJP decoction intervention group (H group, HBJP 20 μ g/mL + TNF- α 10 ng/mL), a HBJP decoction and activator intervention group (H + R group, Recilisib 10 μ M + HBJP 20 μ g/mL + TNF- α 10 ng/mL), an inhibitor intervention group (positive control group, LY294002 10 μ M + TNF- α 10 ng/mL), and a HBJP control group (20 μ g/mL).

2.8. Inflammatory cytokines detection

IL-1 β , IL-2, IL-6 and IL-10 levels of serum in rats and supernatant in IEC-6 were detected by ELISA kit (Solarbio science and technology Co Ltd. Beijing, China) according to the manufacturer's instructions, respectively. Optical density was measured at 450 nm to obtain the absorbance value of samples. Each test has been repeated three times.

2.9. Western blot analysis

Total protein was extracted from colon tissue and IEC-6 using lysis buffer (Thermo Fisher Scientific Inc., Waltham, USA) that contained pre-added protease inhibitors (Solarbio science and technology Co Ltd. Beijing, China). The protein concentration was measured with a BCA kit (Beyotime Biotechnology Co., LTD, Shanghai, China). First, the protein standard in the kit was diluted to obtain the standard curve. Then, the total protein of each group and the working solution were added to a 96-well plate and placed at 37 °C for 30 min. The protein concentration of the samples was calculated by measuring the absorbance at 562 nm. Equal amounts of protein were separated by 10% sodium dodecyl sulfate polyacrylamide gel electrophoresis (Yamei Biotechnology Co., LTD, Shanghai, China), and electro transferred onto polyvinylidene difluoride (PVDF) membranes (Thermo Fisher Scientific, Inc., USA). The membranes were blocked with closing fluid, and incubated with primary antibodies including PI3K (1:1000), P-PI3K (1:1500), Akt (1:1000), P-

Akt (1:1500), NF- κ B (1:1000), P-NF- κ B (1:1500) and GAPDH (1:5000) (Cell Signaling Technology, USA) overnight at 4 °C. Washing with TBST, the membranes were incubated with horseradish peroxidase conjugated secondary antibodies (1:5000) for 1.5 h at room temperature. Finally, the proteins were visualized using an enhanced chemiluminescence detection system (Amersham Biosciences Europe, Germany). All tests were repeated three times.

2.10. Statistical analysis

The SPSS was used for statistical analysis and one-way ANOVA was used for statistical comparison. $P \leq 0.05$ was considered statistically significant.

3. Results

3.1. Components of HBJP decoction

To identify the main chemical constituents, the components of HBJP decoction were measured by UPLC-QE-MS. The total positive and negative ion chromatograms of HBJP decoction are shown in Fig. 1A and B, and the chemical composition of all compounds were showed in supplementary materials (Supplementary Tables S1 and S2), and pulchinoside A may be from radix pulsatillae, epiberberine and berberine may be from rhizoma coptidis, gallic acid may be from radix sanguisorbae, magnolol may be from magnolia officinalis, atractylenolide may be from rhizoma atractylodis macrocephalae, poricoic acid A may be from poria cocos, atractyloside A may be from rhizoma atractylodis, lobetyolin may be from radix codonopsis, ellagic acid may be from pomegranate rind, paeoniflorin may be from white peony root, and liquiritin may be from radix glycyrrhizae. These chemical constituents were preliminarily detected by mass spectrometry using the BIOTREE PWT database of Shanghai BIOTREE Biological Technology Co., Ltd. (Shanghai, China), which may be the active components of HBJP decoction.

HBJP decoction samples were examined by UHPLC-MS/MS. Base peak ion (BPI) chromatogram in positive (A) and negative (B) ion modes for HBJP decoction samples are shown.

3.2. The therapeutic effects of HBJP decoction on diarrheal rats

3.2.1. HBJP decoction reduce diarrhea symptoms

During the course of the experiment, the rats in the control group remained healthy with good mental, clean and lustrous hairs, and normal stools (Fig. 2A and B). At day 17 of modelling test, rats in other groups showed severe diarrhea with sticky and loose feces, poor mental, yellow and lusterless hair (Fig. 2C–F). After being treated for 5 d, the diarrheal scores and semi-quantitative scoring in all HBJP and PD groups were notably decreased (Table 3 and Table 4), and the clinical symptoms in rats were alleviated (Fig. 2G–I). The rats in control group showed gradual weight gain, but the body weight of rats in other groups decreased gradually throughout the modelling process (Fig. 4). After being treated with HBJP decoction, the body weight of rats increased, especially in PD and H-HBJP groups ($P < 0.05$). As shown in Fig. 3, the body temperature of the rats in the control group remained the normal

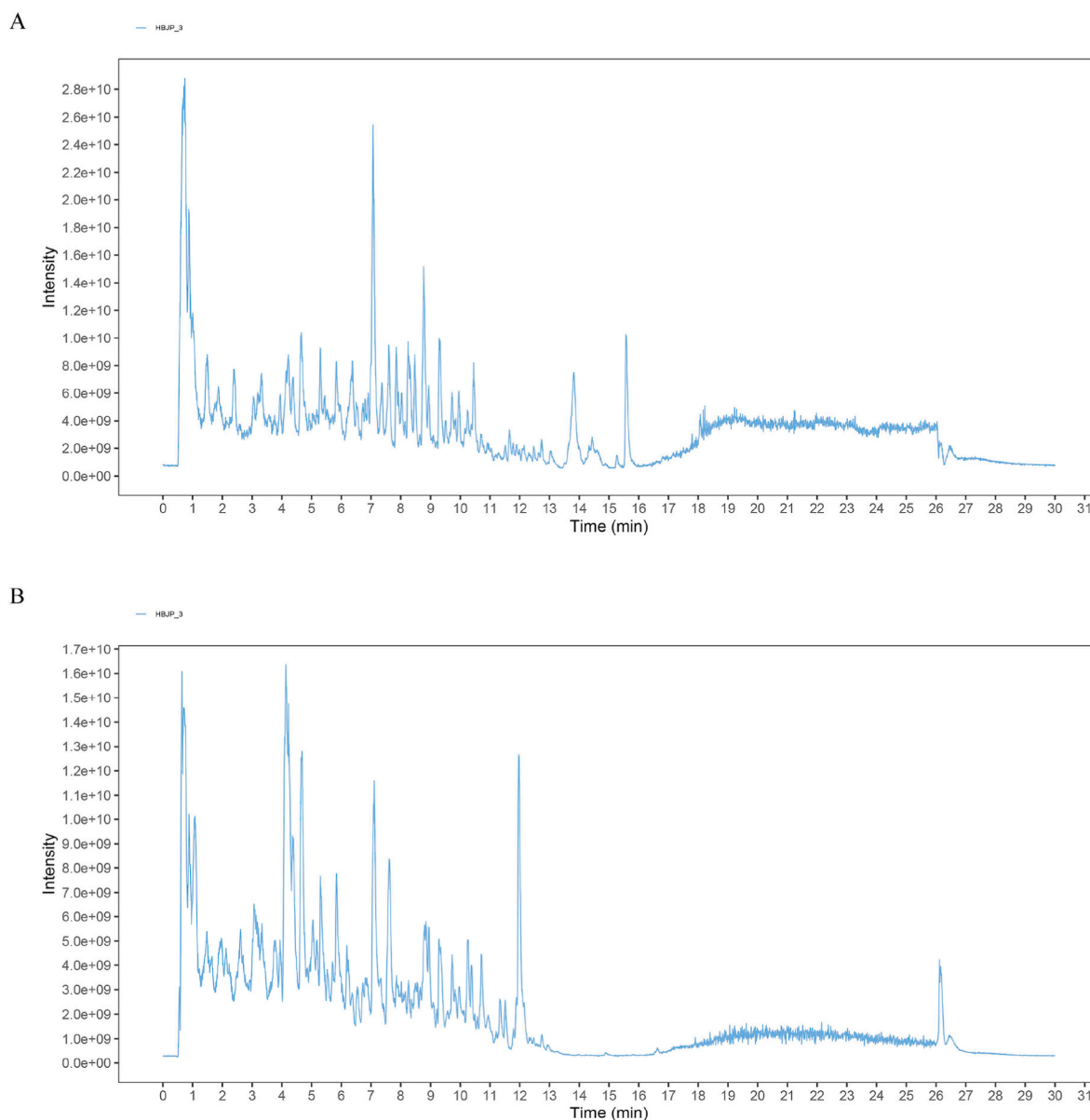


Fig. 1. Identification of chemical components of HBJP decoction.

range, while that of the rats in the other groups increased notably at day 17 of modelling test, and returned to normal levels after being treated with HBJP decoction and PD. These results indicated that mock test on diarrhea in rats were successful, and HBJP decoction and PD could effectively attenuated the clinical symptoms in diarrheal rats.

3.2.2. HBJP decoction improved the abnormal changes in routine blood test

As shown in Fig. 5, WBC, MONO, MPV, and PCT in the model group were notably higher than those in the control group, and these abnormal changes were improved in HBJP groups and PD group. WBC in the M-HBJP, H-HBJP and PD groups were significantly lower than those in model group ($P < 0.05$ or $P < 0.01$), and that in L-HBJP group was decreased ($P > 0.05$). Compared with the model group, MPV and PCT were significantly decreased in all treatment groups ($P < 0.05$). IRF, NEUT, RET, and PLT in model group were significantly lower than those in the control group, and these abnormal indexes were increased in HBJP groups and PD group. HBJP decoction had a dose-dependent regulation effect on blood routine indexes in rats, among which H-HBJP group was better than M-HBJP group, M-HBJP group was better than L-HBJP group, and the effect of H-HBJP group was close to PD

group. The results indicated that inflammatory response, anaemia, and dysfunction of spleen occurred in model rats, and HBJP was able to alleviate these changes in a dose-dependent manner.

3.2.3. Effects of HBJP decoction on blood biochemistry

As shown in Fig. 6, AMY, GLOB, GLU, NA, and BUN in the model group were significantly increased, and the ALB, CA, and PHOS were significantly decreased ($P < 0.01$). Each of treatment groups had regulatory effects on these indexes. Besides BUN in M-HBJP group, other parameters in all treatment groups were improved significantly, and H-HBJP group and PD group had the most obvious effect on these indexes. The results suggested that rats in model group had electrolyte disorder and liver dysfunction, and therapeutic effect of high dose of HBJP on these changes of blood biochemistry was better than middle and low dose of HBJP.

3.2.4. Effects of HBJP decoction on inflammatory cytokines in serum

Inflammatory response has been reported as an important factor in the development of diarrhea (Fischer et al., 2016). ELISA results showed that the expression of IL-1 β , IL-2 and IL-6 were significantly increased,



Fig. 2. The clinical symptoms of rats. A and B showed the state of healthy rats. C, D, E, F showed the state of model rats. (C) Mental state; (D), (E) the loose, sticky and yellow stools, and the foul Crissum and tail; (F) the dark yellow urine. G, H and I showed the state of rats treated with HBJP decoction. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 3

The diarrhea index scores of rats in all groups during the experiment.

Groups	Day 1 (Beginning of molding)	Day 17 (After molding)	Day 23 (After treatment)
Control group	0	0	0
Model group	0	3.12 ± 0.35	/
L-HBJP group	0	2.82 ± 0.46	0.04 ± 0.07##
M-HBJP group	0	2.54 ± 0.35	0.10 ± 0.15##
H-HBJP group	0	3.95 ± 0.32	0.03 ± 0.07##
PD group	0	2.36 ± 0.37	0.02 ± 0.06##

Note: ## presents $P < 0.01$, vs. the 17th day (after modelling).

while IL-10 markedly decreased in mock rats (Fig. 7). The expression levels of pro-inflammatory cytokines IL-1, IL-2 and IL-6 were significantly decreased, and the anti-inflammatory cytokine IL-10 were significantly increased in HBJP decoction groups (Fig. 7). What's more, the intervention effect of H-HBJP group was better than M-HBJP group and L-HBJP group, and was not obviously different from PD group. These results suggested that inflammation was alleviated by HBJP decoction in diarrheal rats.

3.2.5. Histopathological changes of main organs

Histopathological examination revealed no obvious abnormalities in the organs of rats in control group, but lesions in the colon, lungs, and

Table 4

The semi-quantitative scoring of rats in all groups during the experiment.

Groups	Day 1 (Beginning of molding)	Day 10 (During molding)	Day 17 (After molding)	Day 23 (After treatment)
Control group	0	0	0	0
Model group	0	5.67 ± 0.82##	11.00 ± 1.55	/
L-HBJP group	0	6.00 ± 1.10##	10.50 ± 1.52	1.00 ± 0.00##
M-HBJP group	0	5.33 ± 1.63##	9.67 ± 1.37	0.67 ± 0.52##
H-HBJP group	0	5.17 ± 1.72##	9.50 ± 1.38	0.67 ± 0.52##
PD group	0	5.17 ± 1.47##	10.83 ± 1.72	0.67 ± 0.52##

Note: ## presents $P < 0.01$, vs. the 17th day (after modelling).

liver were noted in model group. Integrity of the colonic epithelium was destroyed, and the infiltration of inflammatory cells into colon tissues was observed in model rats (Fig. 8). After administration of HBJP decoction for 5 d, the integrity of the mucosal epithelium in the colon regenerated and recovered gradually. The villi of alveolar spaces were damaged and detached, and the alveolar septa was congested and thickened in model group (Fig. 9). In HBJP groups, the alveolar septa narrowed, congestion was alleviated, and the villi of alveolar spaces gradually recovered with a dose-response relationship. In model group, hepatic cords were disordered, and a large number of red cells were

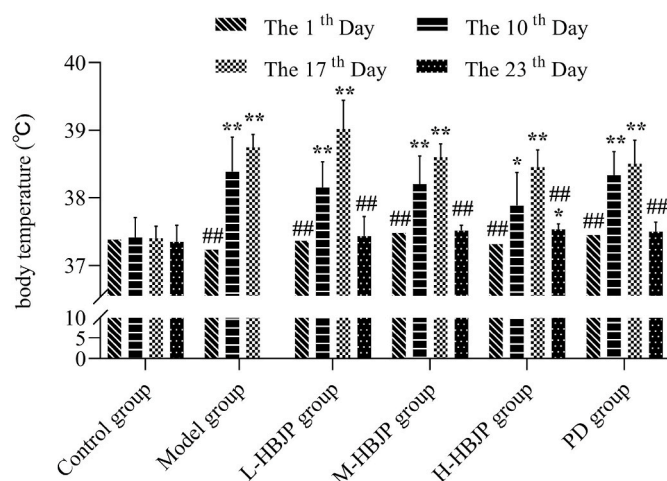


Fig. 3. Body temperature changes of rats in all groups during the experiment. The temperature remains within the normal range in the control group during the whole course of the experiment. The temperature gradually increased in other groups over 17 d (modelling phase), and gradually decreased to the normal range from the 18th to 23rd day of test (treating phase). * presents $P < 0.05$, vs. the first day, # presents $P < 0.05$, vs. the 17th day.

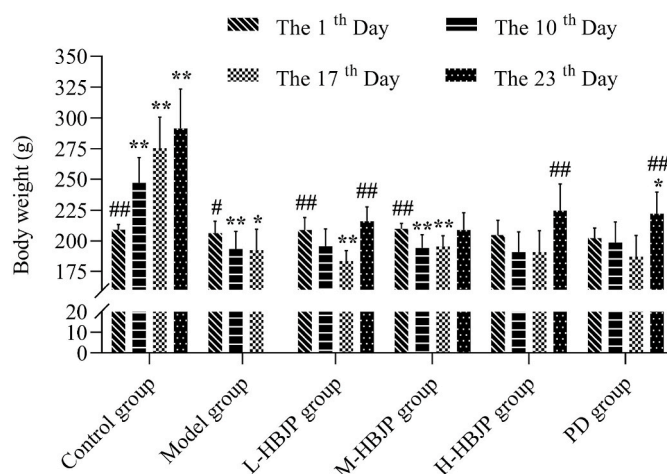


Fig. 4. Body weight changes of rats in all groups during the experiment. The weights increased gradually in the control group during the whole course of the experiment. The weight decreased gradually for other groups over 17 d (modelling phase), and gradually increased from the 18th to 23rd day (treating phase). * presents $P < 0.05$, vs. the first day, # presents $P < 0.05$, vs. the 17th day.

observed in central veins and hepatic sinusoidal spaces, which showed severe congestion, and some hepatocytes showed fatty degeneration (Fig. 10). After being treated with the HBJP decoction, the congestion in central veins and hepatic sinusoidal spaces was strikingly decreased, hepatic cords returned to normal, and hepatocyte steatosis was rarely observed.

3.3. Effect of HBJP decoction on PI3K/Akt/NF- κ B signal pathway in diarrheal rats

PI3K/Akt/NF- κ B signal pathway participates in local inflammation in the intestinal tract. It has been shown that inhibition of PI3K/Akt/NF- κ B signal pathway could reduce inflammatory response (Lu et al., 2018). In this study, we assessed the phosphorylation of PI3K, Akt and NF- κ B in rats. Western Blot analysis showed that the proteins level of P-PI3K, P-Akt and P-NF- κ B were significantly increased in the model group

(Fig. 11). HBJP decoction and PD treatment reduced the level of these proteins with a dose dependent manner (Fig. 11). H-HBJP and PD groups had the best intervention effect than M-HBJP and L-HBJP groups. The data suggested that therapeutic effect of HBJP decoction on diarrhea might be through PI3K/Akt/NF- κ B signal pathway in rats.

3.4. Cell viability detection

To investigate the appropriate concentration of TNF- α for the *in vitro* inflammatory model, and the regulatory effects of HBJP decoction on inflammation, we used these two substances with different concentrations to treat IEC-6, and searched for their appropriate doses for subsequent *in vitro* experiments.

TNF- α inhibited IEC-6 viability in a concentration dependent manner under 10 ng/mL, and the activity of IEC-6 was inhibited at 10 ng/mL ($P < 0.01$) (Fig. 12A). The effect of HBJP decoction on IEC-6 activity was not obvious in the range of 0–20 μ g/mL, but it inhibited IEC-6 activity over 40 μ g/mL ($P < 0.05$) (Fig. 12B). Therefore, we chose 10 ng/mL of TNF- α for the *in vitro* inflammation model stimulation, and 20 μ g/mL of HBJP for the intervention experiment.

3.5. HBJP decoction reduced inflammatory response was depend on PI3K/Akt/NF- κ B signal pathway in the TNF- α stimulated IEC-6

In vivo results suggested that HBJP decoction played a critical role in suppression inflammation in rat diarrhea. In order to investigate the effect of HBJP decoction on inflammatory response *in vitro* test, TNF- α was used to stimulate inflammatory response in IEC-6 according to reference (Nie et al., 2020). Compared to control group, the expression of IL-1 β , IL-2 and IL-6 was upregulated and the expression of IL-10 was downregulated in TNF- α treated groups. As shown in Fig. 13, HBJP decoction treatment was efficiently inhibited the expression of IL-1 β , IL-2 and IL-6, and increased the expression of IL-10.

To further explore the role of the PI3K/Akt/NF- κ B signal pathway in HBJP decoction alleviating inflammatory response, the phosphorylation of related proteins in this signal pathway was detected by Western blot analysis. After being treated with TNF- α for 24 h, the levels of P-PI3K, P-Akt and P-NF- κ B were increased in IEC-6 ($P < 0.05$). HBJP decoction clearly suppressed expression upregulation of P-PI3K, P-Akt and P-NF- κ B in IEC-6 induced by TNF- α (Fig. 14). Similarly, LY294002, a broad-spectrum inhibitor on PI3K, markedly attenuated the expression of these proteins. Recilisib, an activator of Akt and PI3K, blocked the regulation of HBJP decoction on the levels of P-PI3K, P-Akt and P-NF- κ B. Taken together, these results suggested that HBJP decoction alleviated inflammatory response induced by TNF- α in IEC-6, which was associated with the regulation of PI3K/Akt/NF- κ B signal pathway.

4. Discussion

As a common disease in calves, diarrhea has always been problematic for the cattle industry. Because the efficacy of antibiotics to treat this disease is declining, it is particularly important to develop some complementary and alternative medicine. The HBJP decoction has been shown to be effective for treating calf diarrhea. This decoction is a kind of complementary and alternative medicine for diarrhea in calves. In this study, the typical compounds of 11 herbs in HBJP decoction were identified by UPLC-QE-MS analysis. The diarrheal model in rats was successfully established, and the diarrheal symptoms were ameliorated by HBJP decoction. The concentration of pro-inflammatory factors in serum and the levels of P-PI3K, P-Akt and P-NF- κ B proteins in colon of diarrheal rats were notably increased, which were adjusted by HBJP decoction. The similar effects of HBJP decoction were observed *in vitro* test. Therefore, to inhibit the PI3K/Akt/NF- κ B signal pathway might be a vital target of HBJP decoction for treating diarrhea in clinical practice.

The HBJP decoction is composed of eleven herbs. The chemical constituents of HBJP decoction were preliminarily detected by UPLC-QE-

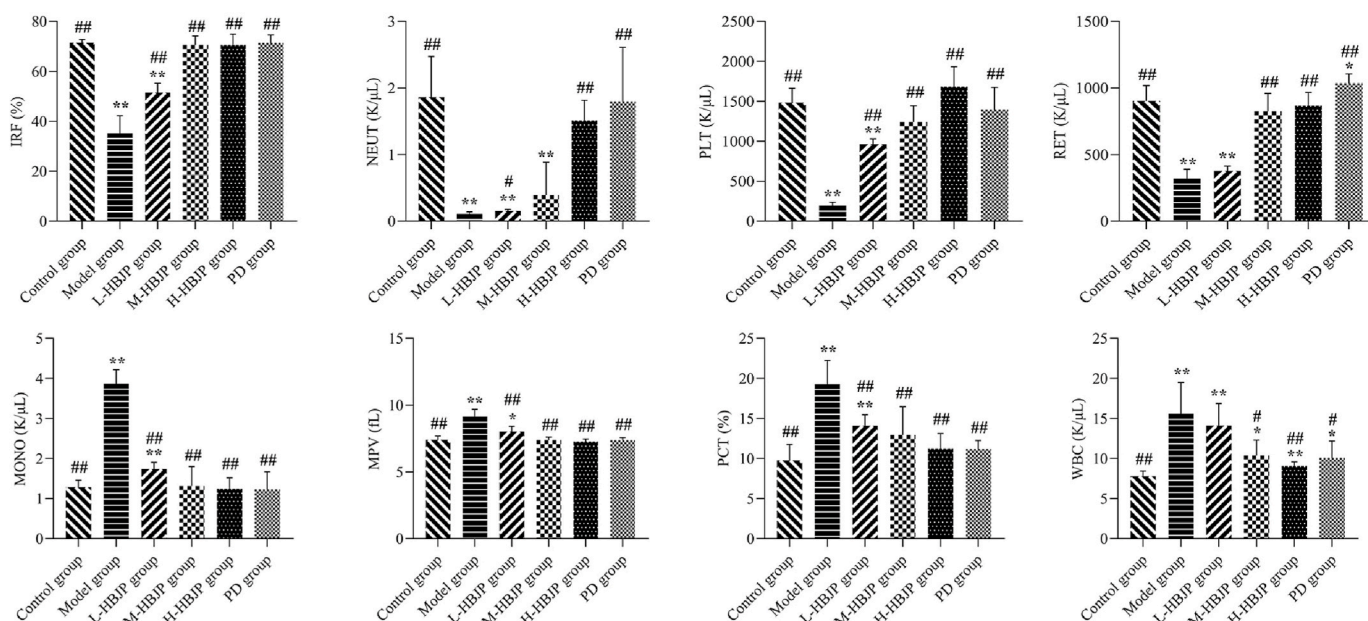


Fig. 5. Results of blood routine. # represents a significant difference compared to the model group, and * represents a significant difference compared to the control group ($P < 0.05$).

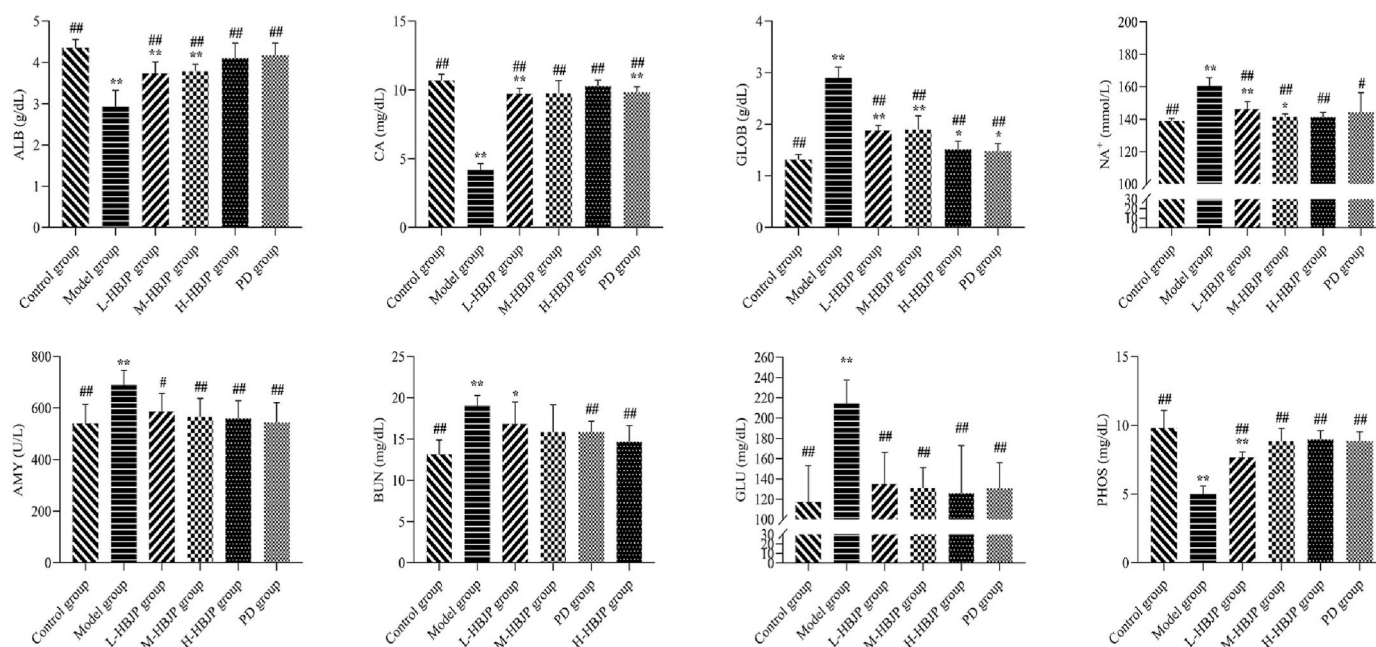


Fig. 6. Results of blood biochemical indexes. # represents a significant difference compared to the model group, and * represents a significant difference compared to the control group ($P < 0.05$).

MS, which is increasingly used in the detection of chemical components of the traditional Chinese medicine (Xuan et al., 2021). In components of HBJP decoction, Atractylenolide, Paeoniflorin, Liquiritigenin and Magnolol are known to have anti-inflammatory activities (Chiu et al., 2021; Huang et al., 2019; Ji et al., 2016; Zhou et al., 2020). Berberine, gallic acid and magnolol can effectively treat diarrhea and reduce intestinal damage (Lescano et al., 2016; Xia et al., 2019; Zhang et al., 2021). Paeoniflorin and ellagic acid have a regulatory effect on intestinal flora (Fan et al., 2020; Qi et al., 2022). In this study, we explored the anti-diarrheal effects of HBJP decoction and its underlying molecular mechanisms. These compounds may be potential components providing the therapeutic effects of HBJP decoction in diarrhea. Therefore, further

study of the therapeutic mechanism of this medicine is important.

We explored the mechanisms behind HBJP decoction treatment in diarrheal rats. Blood biochemistry assays showed that calcium level of serum in rats with diarrhea was significantly decreased, and serum sodium level was significantly increased, which suggested that an electrolyte imbalance occurred during progress of the diarrheal episode. The notable decrease of ALB in the model group indicated that the metabolism and absorption of nutrients may be abnormal during the pathogenesis of diarrhea, thus affecting growth performance (Zafar and Khan, 2019). These results coincided with a decrease in body weight of rats in the model group. AMY is mainly secreted by the salivary glands and pancreas, and its level is increased by gastrointestinal diseases and liver

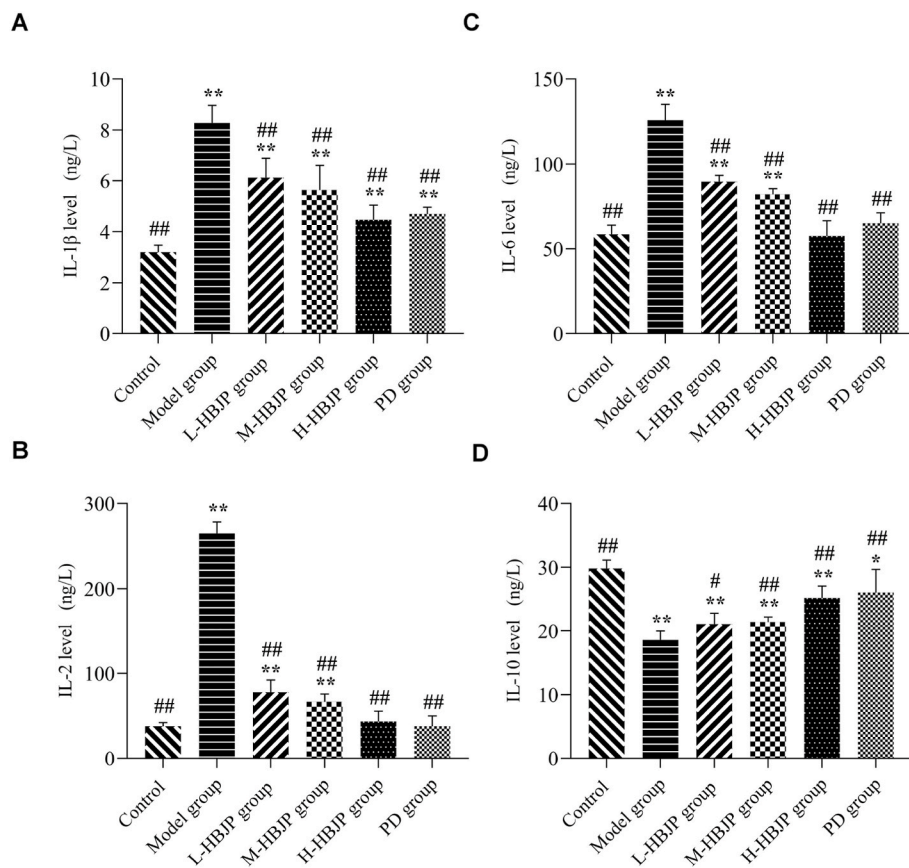


Fig. 7. The contents of inflammatory cytokines in serum. (A) Interleukin (IL)-1 β , (B) IL-2, (C) IL-6, (D) IL-10. The contents of IL-1 β , IL-2, IL-6 and IL-10 in the model group were significantly different from those in the control group ($P < 0.01$), and the content of IL-1 β , IL-2 and IL-6 in the H-HBJP, M-HBJP and L-HBJP groups were significantly lower than that in the model group ($P < 0.01$). The level of IL-10 in HBJP group was higher than that in model group, and the increase of IL-10 in L-HBJP group was not significant, while that in M-HBJP and H-HBJP groups was significant ($P < 0.05$, $P < 0.01$). # represents a significant difference compared to the model group, and * represents a significant difference compared to the control group ($P < 0.05$).

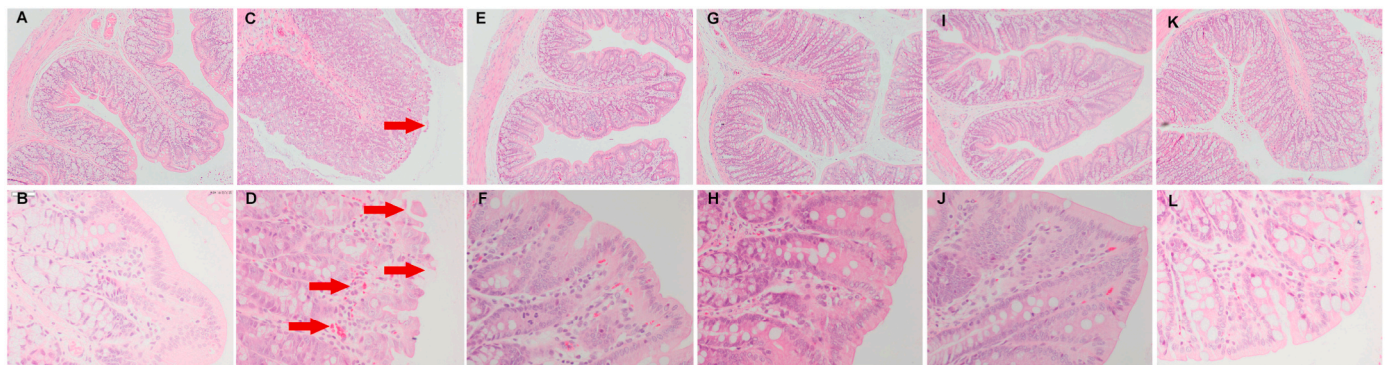


Fig. 8. Histopathological changes of the colon. Control group (A, B), the epithelium was intact, and the brush border was clearly visible. Model group (C, D), the epithelium was destroyed, and a lot of red cells were observed in the veins and arteries, suggesting serious congestion. HBJP group (E, F, G, H, I, J), the mucosal epithelium regenerated, the integrity was recovered and congestion was effectively alleviated. PD group (K, L) were similar to those of HBJP group. Arrows indicate the injuries of mucosal epithelium, severe congestion, respectively. Original magnification for A, C, E, G, I, K was 100 $\times$; Original magnification for B, D, F, H, J, L was 400 $\times$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

diseases besides pancreatic disease (Hua et al., 2020). Here, we found that the levels of GLOB and AMY increased in the serum of diarrheal rats, which might be related to the associated inflammation or dehydration. In addition, the increase of serum GLU may have been related to drinking honey water and liquor over a long period. After treatment with HBJP decoction, a notable difference was found in these blood biochemical indexes. Therefore, we concluded that the HBJP decoction could treat diarrhea by regulating the levels of electrolytes and proteins.

WBC and PCT are blood biochemical markers associated with inflammation. Both markers were significantly increased in the model group, and the ratio of MONO to NEUT was seriously disturbed, which indicated that there was an acute inflammatory response in the diarrheal

rats. NEUT are the main phagocytes constituting the front line of innate immune defence against bacterial infection. It is possible that alcohol could injure hematopoietic tissue, thus impairing granulocytes (Shi et al., 2019). Furthermore, a high-fat diet decreased a marker of neutrophil activation in mice (Cavallari et al., 2016). The decrease of neutrophils and the changes of other immune cells may be related to the influence of these factors on the immune function of rats during modelling process. The inflammatory response was also reflected in the pathological changes of some organs such as the infiltration of inflammatory cells in the colon, lung, and liver. The notable decreases in IRF, RET, and PLT indicated that the model rats may have had an accompanying blood system injury. Alcohol consumption can impair host

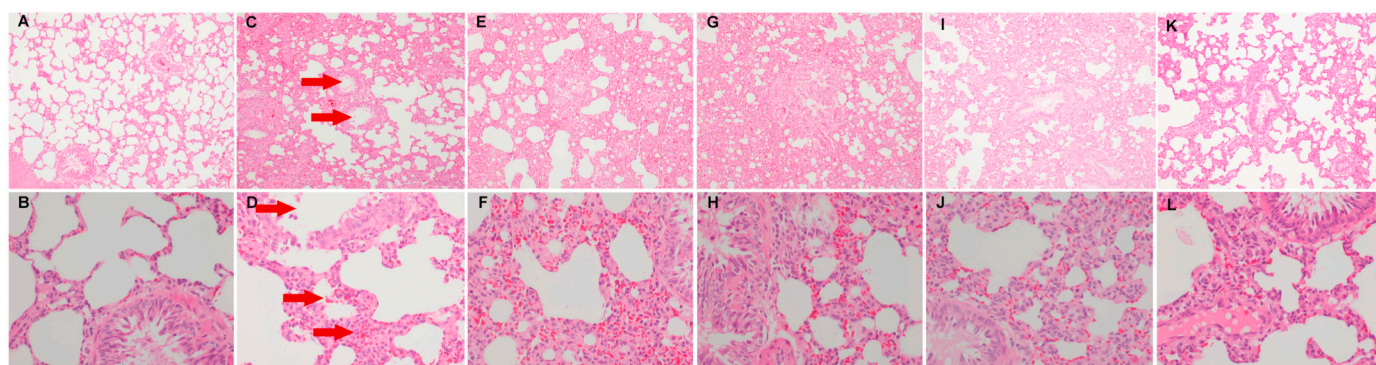


Fig. 9. Histopathological changes of the lung. Control group (A, B), the morphologies of alveolar septa and alveolar spaces were normal. Model group (C, D), the villi of alveolar spaces fell off and damaged, the alveolar septa were congested and thickened. HBJP groups (E, F, G, H, I, J), the alveolar septa became narrow, the congestion was alleviated and the villi of alveolar spaces gradually recovered. The effect is dose dependent. PD group (K, L), the results were similar to those of H-HBJP group. Arrows indicate damaged villi of alveolar spaces, congestion and the widened alveolar septa. Original magnification for A, C, E, G, I, K $\times 100$; Original magnification for B, D, F, H, J, L $\times 400$.

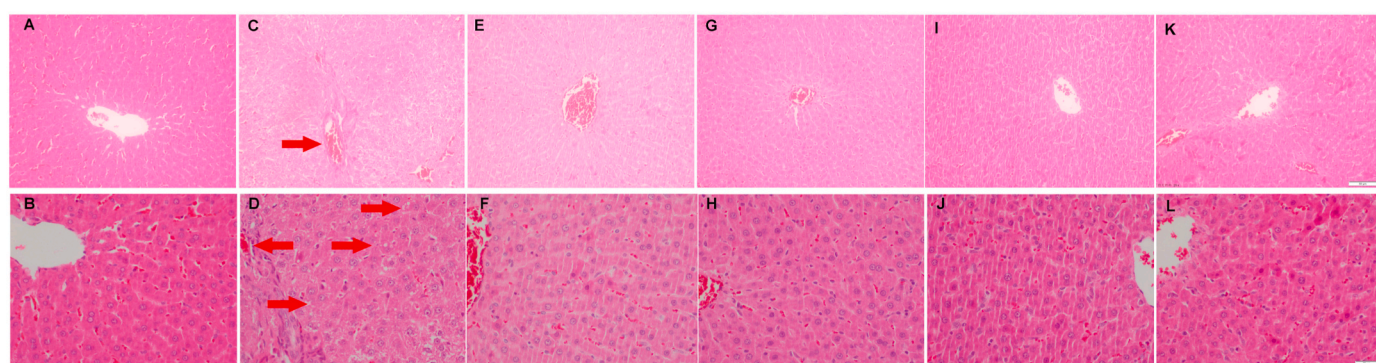


Fig. 10. Histopathological changes of the liver. Control group (A, B), hepatic cords were orderly, red cells were not observed in the central veins and hepatic sinusoidal spaces, and the morphologies of hepatocytes were normal. Model group (C, D), hepatic cords were disordered, a large number of red cells were observed in central veins and hepatic sinusoidal spaces, suggesting severe congestion, and some hepatocytes showed fatty degeneration. HBJP groups (E, F, G, H, I, J), the congestion in central veins and hepatic sinusoidal spaces strikingly decreased, the hepatic cords were returned to normal, and hepatocytes steatosis were rarely seen. PD group (K, L), the results were similar to those of H-HBJP group. Arrows indicate the congestion in central veins, fatty degeneration of hepatocytes, and disordered hepatic cords. Original magnification for A, C, E, G, I, K $\times 200$; Original magnification for B, D, F, H, J, L $\times 400$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

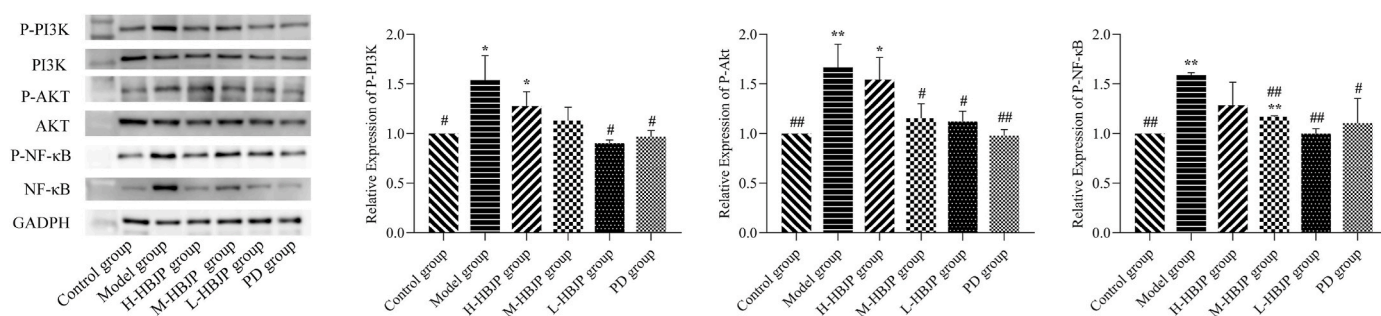


Fig. 11. The effects of HBJP decoction on the relative protein levels of phosphorylated PI3K/AKT/NF- κ B. For each indicator, # represents a significant difference compared to the model group and * represents a significant difference compared to the control group ($P < 0.05$).

immunity and incur thrombocytopenia (Silczuk and Habrat, 2020). Intestinal injury caused by *E. coli* in rats may also be responsible for the decrease in PLT. Various effective components of HBJP decoction have anti-inflammatory (Chiu et al., 2021; Du et al., 2019; Kang et al., 2019) and therapeutic effects on blood diseases such as anemia and vascular dysfunction (Ku et al., 2017; Ong et al., 2018). Therefore, these blood biochemical indexes in the rats recovered after HBJP treatment.

Histopathological tests showed that lesions were the most serious

manifestation in colon, and the integrity of colon was destroyed. In addition, the pathological changes occurred in the liver and lungs. These changes were responsible for the changes of blood indexes such as WBC, NEUT, GLOB, ALB, and BUN. In diarrheal cases in animals, the dysfunction of the large intestine is destroyed usually, and the lungs share an interior-exterior relationship with the large intestine in TCM, which could be one of causes in lung lesions (Hua et al., 2020). Liver disorder could have resulted from the high-sugar-high-fat diet and

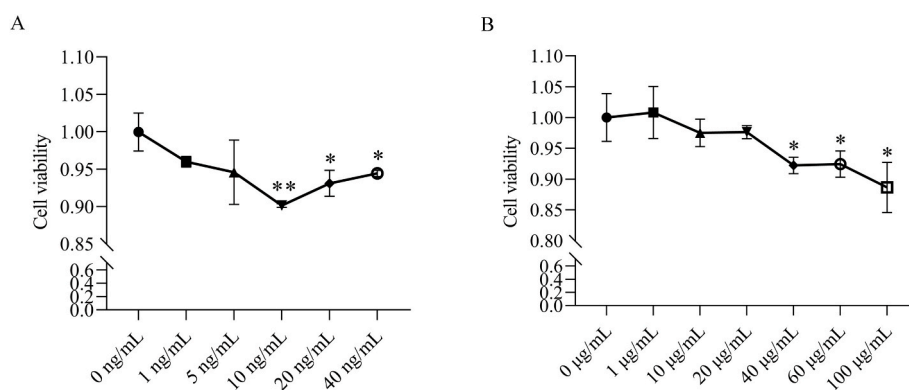


Fig. 12. The viability of IEC-6. A, the effect of TNF- α on IEC-6 viability. B, the effect of HBJP decoction on IEC-6 viability.

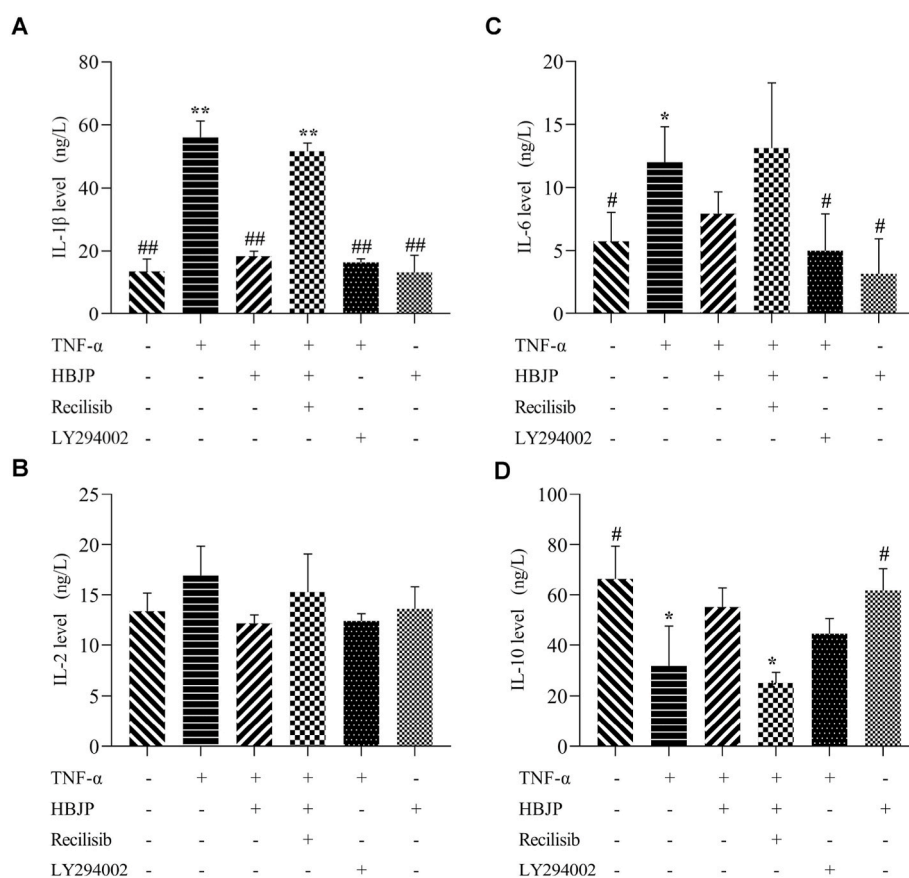


Fig. 13. The levels of inflammatory cytokines in IEC-6. For each indicator, * represents a significant difference compared to the control group and # represents a significant difference compared to the model group.

alcohol (Silva et al., 2021). The HBJP decoction alleviated organ injuries of rats in the model group. Studies showed that white peony root, radix glycyrrhizae, cortex magnoliae officinalis, poria and their active components had hepatoprotection effects (Chen et al., 2020; Yimam et al., 2016; Zhang et al., 2014); Radix pulsatillae, rhizoma atractylodis macrocephalae and radix glycyrrhizae suppress lung lesions (Liu et al., 2019; Yuan et al., 2020); Pomegranate rind and radix sanguisorbae protect the intestinal mucosa (Al-Gubory et al., 2021; Yasueda et al., 2020). Therefore, HBJP decoction showed comprehensive regulation and protection on main organs in diarrheal rats.

Diarrhea is a gastrointestinal disease that is commonly accompanied with inflammation, and the expression of some inflammatory markers is used to evaluate the severity of this disease (Wong et al., 2021). In the

present study, the contents of serum pro-inflammatory cytokines including IL-1 β , IL-2, and IL-6 increased, and anti-inflammatory cytokine IL-10 decreased in diarrheal rats, which is consistent with results of the previous study (Yao et al., 2017). The infiltration of the inflammatory cells and the injury of the liver, lungs, and colon are promoted by the inflammatory responses. In addition, the notable increase in WBC and MONO also indicated that model rats endured an acute inflammatory response. After treatment with the HBJP decoction, the inflammatory cytokines and pathological changes in diseased rats restored. This regulation of HBJP decoction on the cytokines was verified again in the inflammation model on IEC-6 stimulated by TNF- α . Some evidences showed that radix pulsatillae, rhizoma coptidis, radix sanguisorbae, pomegranate rind and magnolia officinalis have good anti-inflammatory

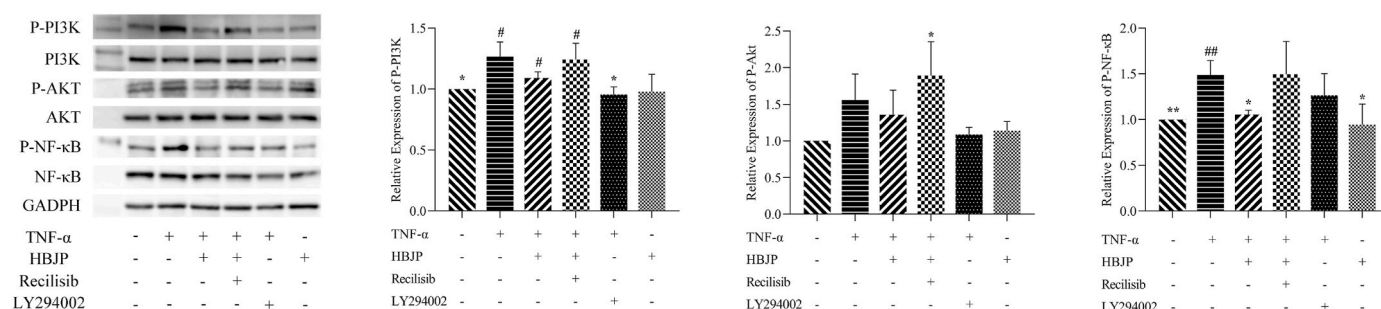


Fig. 14. The effects of HBJP decoction on the relative protein of PI3K/Akt/NF- κ B. For each indicator, # represents a significant difference compared to the control group and * represents a significant difference compared to the model group ($P < 0.05$).

activity (Chiu et al., 2021; Du et al., 2019; Fujii et al., 2017). All these data demonstrated that HBJP decoction could inhibit the inflammatory response in animal diarrhea that is caused by multiple aspects.

We further explored the mechanism of HBJP decoction anti-inflammation in diarrheal rats. Many pathways typically involved downstream of PI3K/Akt contribute to maintaining and amplifying inflammation, thereby exacerbating the disease process. Typically, NF- κ B can be activated by the PI3K/Akt signal pathway, to upregulate many inflammatory factors such as IL-1, IL-2, IL-6, and TNF- α , which are often associated with diarrheal processes and may promote the damage caused by disease (Wu et al., 2013). In this study, the expression levels of P-PI3K, P-Akt and P-NF- κ B increased notably in the model group, and the expression levels of phosphorylated PI3K, Akt, and NF- κ B were inhibited or attenuated by HBJP decoction, which has also been observed in inflammatory IEC-6 caused by TNF- α . Therefore, HBJP decoction may reduce inflammation in diarrheal rats by diminishing the PI3K/Akt/NF- κ B signal pathway.

There were some limitations in the present study. The chemical composition of the HBJP decoction is complex, and shows multi-anti-inflammatory mechanisms in the treatment of diarrhea. Therefore, the mechanism of HBJP decoction in the treatment of diarrhea still requires further study. In addition, for ethical reasons, we did not set up an untreated group, which could confound the effect of time on treatment. However, Yao et al. found that self-healing rats in a similar modelling method would not fully return to normal levels (Yao et al., 2017). According to both this data and the published evidences, we concluded that HBJP decoction has a therapeutic effect on diarrheal rats.

5. Conclusions

HBJP decoction had a good therapeutic effect on diarrheal rats with a significant dose-response manner. The *in vitro* and *in vivo* studies demonstrated that HBJP decoction can reduce inflammatory response via regulating PI3K/Akt/NF- κ B signal pathway. Altogether, HBJP decoction is considered as a complementary and alternative medicine in the treatment of diarrhea. This data provides an important theoretical basis for further development of new Chinese medicine for the treatment of animal diarrhea.

Ethics approval and consent to participate

The animal study was reviewed and approved by the Laboratory Animal Ethics Commission of the Lanzhou Institute of Husbandry and Pharmaceutical Sciences of CAAS (SYXK [Gan] 2019-0002).

Credit Author Statements

Jianxi Li and Xiaoping Song conceived and designed the work.
Jianxi Li coordinated technical support and funding.
Zunxinag Yan wrote the manuscript.
Zunxiang Yan, Kang Zang, Kai Zhang, Guibo Wang, Lei Wang,

Jingyan Zhang, Zengying Qiu, Zhiting Guo and Yandong Kang performed the experiments and collected the samples.

Jianxi Li and Kai Zhang reviewed the manuscript.

All authors have read and approved to the published version of the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2022.115212>.

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